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A Data-Centric Method to Identify Lung Cancer Using a Variety of Machine Learning Approaches

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***Abstract*— Lung cancer continues to be the leading cause of cancer-related deaths worldwide, and early detection is essential to extending patient life expectancy. The goal of this study is to boost the precision of lung cancer diagnosis through the use of modern machine learning approaches. We developed a thorough data pretreatment strategy that included feature selection, encoding, oversampling, and hyper parameter tuning to handle data discrepancies. We used the Kaggle dataset to assess six ML algorithms: Naive Bayes, Decision Tree, Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), and Logistic Regression. With an accuracy of 99.38%, precision of 99.39%, recall of 99.38%, and F1 score of 99.38%, the SVM approach performed exceptionally well.**

Keywords*— *Lung Cancer, Medical Diagnostic, Prediction, Machine Learning, SVM, Oversampling.

I. INTRODUCTION

One kind of tumor that usually starts inside the tissues surrounding the airways of the lungs is lung cancer. Although tobacco is the main cause, contaminants such as breathing in asbestos and radon gas exposures may trigger lung cancer in nonsmoking [**1**]. Globally, approximately 1.8 million people die each year due to lung cancer [**2**]. Tobacco use is a primary factor in almost 90% of instances [**3**], and there is unquestionably a link between the risk of developing lung cancer and the severity and length of tobacco dependence. Throughout all of their lives, men face a 1 in 15 probability of getting lung cancer, whereas women face a 1 in 17 chance. Such probabilities are substantially greater for smokers [**4**]. In America, the mean lifespan is just 17.7%, while the mortality rates for men and women is 29.5% and 26.1%, respectively [**5**]. Areas which have elevated prevalence of smoking and industrial contaminants have the greatest instances of lung cancer deaths and incidences. This covers a portion of North America, Eastern Europe, and Asia [2]. Immunotherapy continues to demonstrate encouraging outcomes for lung cancer patients [**6**].

Cutting-edge AI, particularly machine learning, plays a crucial role in the early detection of lung cancer. AI systems may detect precursors of cancer by evaluating large amounts of data using CT scans and other medical imaging, frequently by accuracy comparable to that of experienced radiologists. The capacity to detect cancer at a prior to that, more manageable phase is crucial for prompt action, which greatly improves the results of treatments [**7**]. Additionally, AI improves the precision of pulmonary tumor diagnosis. Deep learning models, trained on extensive image datasets, can effectively distinguish between benign and malignant nodules. With this accuracy, fewer needless procedures are performed, critical cases are treated promptly, and diverse data types are integrated to customize diagnostic strategies [**8**]. Utilizing methods to evaluate aggressiveness of tumors and longevity of patients prospects, AI additionally anticipates the course of lung cancer. Customized therapy protocols are informed by such predicting capabilities, which helps physicians make optimal treatment decisions [**9**].

AI helps in medical devices by customizing therapy protocols for patients and enhancing the precision of prognostic and diagnostic evaluations. This may be essential for giving targeted therapy at an early stage and possibly enhancing the results for patients [**10**]. AI's role in lung cancer research is expanding beyond detection and diagnosis to include prognostics and personalized medicine, leveraging complex algorithms to predict patient outcomes and optimize treatment paths [**11**]. However, relying solely on classification and machine learning techniques may not entirely address the issue. It is crucial to develop a robust information-gathering framework and ensure that all necessary prerequisites and criteria are met**.**

- Comprehensive Data Preprocessing: In order to ensure an adequate dataset for machine learning evaluation, this study included extensive data methods for preparation, such as feature selection, data encoding for categorization, and oversampling to correct class inequities.

- Extensive Algorithm Evaluation: Six machine learning algorithms were evaluated-Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), Logistic Regression, Naive Bayes, and Decision Tree-providing a comprehensive comparison of their performance in lung cancer detection.

- Superior Performance of SVM: SVM outperformed the other models under evaluation in terms efficiency measures, demonstrating its efficacy in lung cancer diagnosis with accuracy of 99.38%, precision of 99.39%, recall of 99.38%, and F1 score of 99.38%.

- Detailed Model Evaluation: The research employed many evaluation variables, such as accuracy, precision, recall, F1 score, ROC curve, and confusion matrix, to conduct a comprehensive analysis of each model's performance and guarantee a comprehensive comprehension of their analytical skills.

There are four sections to this research as well. A summary of previous research done in the field is provided in the **Literature Review** section. A detailed explanation of the

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information as well as the procedures employed in this investigation can be found in the **Materials and Methods** section. An evaluation of the experimental data can be found in the section on **Result Analysis.** The **Conclusion** provides a summary of the main findings and recommendations for future research.

II. Literature Review

The research, which was actually carried out by multiple researchers in an identical manner, is displayed here. They conducted additional research on lung cancer classification as a result of the study's findings. Olatunji, S. O. and associates, 2022 [**12**]. The aim of the endeavor is to use common diagnostic elements from the "data world" website to diagnose lung cancer in a preemptive manner. A number of classifiers have been applied during the investigation: Logistic Regression (LR), K-Nearest Neighbor (K-NN), and Support Vector Machine (SVM). Another technique used to determine which set of traits was most useful was Recurring Features Reduction. The outcomes demonstrated that SVM performed well when fifteen features were used. Its recall rate was 98.33%, its precision rate was 96.72%, and its accuracy rate was 97.27%. Patra, R. [**13**], Radial basis function networks, neural networks, support vector machines, random forests, logistic regression, J48, naive Bayes, and K-nearest neighbors were among the methods that the investigator tested for anticipating lung cancer. They demonstrated that, when applied to the lung cancer data, the radial basis function network produced a greater accuracy of 81.25%. E. Dritsas et al. (2022) [**14**]. The purpose of the study is to use online data to compile general medical and description criteria in order to make a preventive detection of lung cancer. Using machine learning (ML) approaches, they created efficient models for identifying individuals who are more likely to acquire lung cancer and, as a result, for starting early therapy to avoid major effects. The study suggests the Rotation Forest, which functions effectively and is evaluated using accepted metrics like accuracy, precision, recall, F-Measure, and area beneath the curve (AUC). The trial evaluation findings showed that the recommended model performed best, with accuracy, precision, recall, and AUC values of 97.1%, 99.3%, and F-Measure, respectively. Associated with Ahmad S. B. (2024) [**15**]. Their objective is to use machine learning to develop extremely reliable models that can identify lung cancer earlier enough. Specialists might discover a model such as this to be quite useful in determining about whether a patient needs a standard assessment or one that is more intense. Doctors can minimize costly and unnecessary therapies by tailoring a treatment strategy based on accurate forecasts, potentially leading to a significant decrease in treatment expenses. Their goal is to develop a long-lasting model that can accurately predict the sickness. They discovered that XGBoost outperformed earlier models based on their research, achieving an amazing accuracy of 96.92%. Furthermore, the accuracies of LightGBM, AdaBoost, Logistic Regression, and Support Vector Machine were 93.50%, 92.32%, 67.41%, and 88.02%, in that order. Faisal et al. (2018) [**16**]. The aim of this investigation conducted by the authors was to examine the biased nature of various algorithms in order to enhance the efficacy of identifying lung cancer. The study involves evaluating models such as Naive Bayes (NB), Support Vector Machine (SVM), and Decision trees using an established set of data that was acquired from the UCI repository. Once the results of individual and ensemble classifiers was contrasted with renowned ensemble techniques, the gradient-boosted Tree outperformed them, achieving an accuracy of 90%. A. Celik et al. (2022) [**17**]. Plenty of researchers, especially in countries with low or intermediate incomes, are using the enormous quantity of cancer-associated data that was collected as a consequence of the accelerated advances in innovation and the digitization of medical care to reduce the cost of tests that are applied to forecast different types of cancer. The study uses a range of machine learning techniques to predict lung cancer. The random forest performed better, with an accuracy of 96.08%, according to the study's findings. It is widely acknowledged that conventional approaches to machine learning, such as SVM and GMM, achieved remarkably well in classification, even though deep learning has primarily taken over the field of analyzing clinical images [**5**]. This specific study reports that SVM outperforms other methods for categorizing with scores of 99.38%, 99.39%, 99.38%, and 99.38% in terms of accuracy, precision, recall, and f1-score, respectively. The following article will demonstrate its treatment.

III. Materials and Methods

Stages of the proposed lung cancer forecasting technique include feature selection, data preprocessing, data sampling, and classification. SelectKBest was utilized to identify the top 12 attributes that facilitate crucial computations. Throughout the initial step of data processing, thorough cleaning techniques and meticulous filtering are employed. To accurately predict lung cancer, Hyperparameter Tuning is used with six categorization algorithms. This detailed method was specifically designed to accurately detect and classify cases of lung disease within the dataset, which contains extensive health status data from 308 patients. The approach and proposed diagnostic process for identifying lung diseases are clearly illustrated in a graphical format, as depicted in **Fig 1**. The image demonstrates how various machine learning algorithms are utilized during the training phase to identify the most effective one. Based on accuracy measures, the Support Vector Machine (SVM) classification model proved to be the best-performing model in the suggested strategy, with a 99.38% classification accuracy.

A. Dataset

The dataset used in this thorough examination of contemporary techniques came through a public web archive that was especially designed with lung disease sufferers in mind. The collection includes comprehensive data from 308 patient records, covering 16 different variables associated with lung cancer [**18**]. Gender, age, smoking, yellow fingers, anxiety, peer pressure, chronic illness, exhaustion, allergy, wheezing, alcohol use, coughing, dyspnea, difficulty swallowing, chest discomfort, and lung cancer are among the aspects of the dataset. Twelve of these 16 qualities were deemed to be the most crucial and were chosen for more research with the aid of SelectKBest. The top 12 characteristics are age, yellow fingers, anxiety, peer pressure, chronic illness, exhaustion, allergy, wheezing, drinking alcohol, coughing, difficulty swallowing, and chest pain. The results of the examination of the data of age shown in **Fig 2,** exhibits, the age group of 56 to 72 years old had the highest prevalence of lung cancer. A closer look at the dataset revealed a number of typical signs amongst those suffering from lung cancer. Particularly, a sizable percentage of the afflicted population regularly expressed symptoms including yellow_fingers, fatigue [**fig 3**], allergy, wheezing, alcohol consuming, coughing [**fig 3**], and chest pain.

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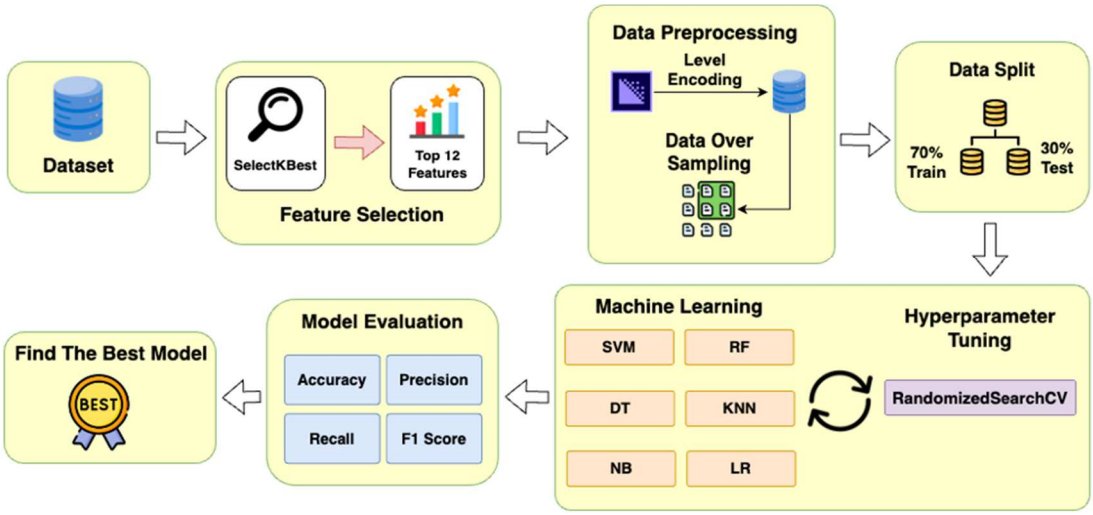


Fig. 1: The suggested methodology's operational process.

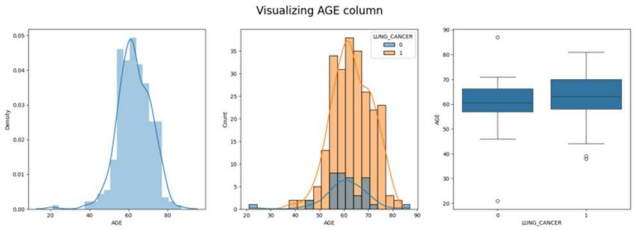


Fig 2: Risk of affection to lung cancer at age

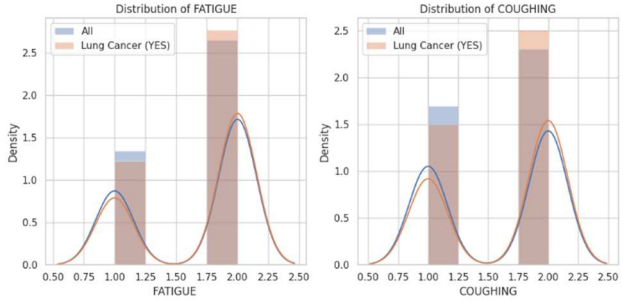


Fig 3: Comparison of Fatigue, Coughing and Shortness of breath between all patients and Lung Cancer positive patients

B. *Data Preprocessing*

In order to adequately prepare the dataset for use in machine learning algorithms, a number of rigorous preprocessing activities have been carried out. Notably, the AGE and LUNG_CANCER features' original forms were in string format. These values in strings were converted into binary values to aid in computation. A binary classification strategy was used because of the diversity in the age range within the dataset. Ages under the average (62.909) were encoded as 0, and ages at or above the average (62.909) were encoded as 1. This methodology guaranteed uniformity and enabled significant cross-age group comparisons. As this was going on, other features in the dataset that were initially coded as 1 or 2, denoting false or true states, were equally encoded as 0 or 1. The dataset was made more manageable and ready for incorporation into machine learning classifiers with this modification. A clear and succinct summary of the data transformations used is given by the comprehensive explanations of these preparation procedures and the accompanying summary table (**Table I**). These preparation steps played a vital role in improving the dataset's appropriateness for further machine learning evaluations, which sought to forecast lung cancer outcomes with increased precision and dependability. Additionally, the dataset is balanced by the usage of Random Over Sampler. Unbalanced statistics are addressed with Random Over Sampler. According to this investigation, there appears to be an uneven distribution of classes in the target variable LUNG_CANCER (more instances of "No" than "Yes"). Oversampling is a strategy used to equalize the dataset by synthesizing artificial samples of the minority class, which in this case is probably the 'Yes' class for lung cancer. This helps prevent the algorithm from having bias in favor of the larger class and improves its ability to learn patterns from the less frequent class. After the oversampling process, 70% of the information were used for training and the remaining 30% for testing. The training dataset consisted of 378, 12 occurrences, while the test dataset had 162, 12 instances. This indicates that the oversampled data frame's overall shape is (539, 12). Now that the data has been suitably divided, it is ready to be assessed with several machine learning classifiers.

TABLE I. DATA ENCODING INFORMATION FOR EACH FEATURE

Attribute	Value	Processed Value
gender	(0 - 1) 0 = Female, 1 = Male	(0 - 1)
age	(21 - 87) Average = 62.909	(0 - 1) 0 = Age < Average, 1 = Age > Average
yellow_fingers	(1 - 2)	(0 - 1) 1 converted to 0 and 2 converted to 1
anxiety	(1 - 2)	(0 - 1) 1 converted to 0 and 2 converted to 1
peer_pressure	(1 - 2)	(0 - 1) 1 converted to 0 and 2 converted to 1
chronic disease	(1 - 2)	(0 - 1) 1 converted to 0 and 2 converted to 1
fatigue	(1 - 2)	(0 - 1) 1 converted to 0 and 2 converted to 1
allergy	(1 - 2)	(0 - 1) 1 converted to 0 and 2 converted to 1
wheezing	(1 - 2)	(0 - 1) 1 converted to 0 and 2 converted to 1

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alcohol consuming	(1 - 2)	(0 - 1) 1 converted to 0 and 2 converted to 1
coughing	(1 - 2)	(0 - 1) 1 converted to 0 and 2 converted to 1
swallowing difficulty	(1 - 2)	(0 - 1) 1 converted to 0 and 2 converted to 1
chest pain	(1 - 2)	(0 - 1) 1 converted to 0 and 2 converted to 1
lung_cancer	(0 – 1) 0 = No, 1 = Yes	(0 – 1)

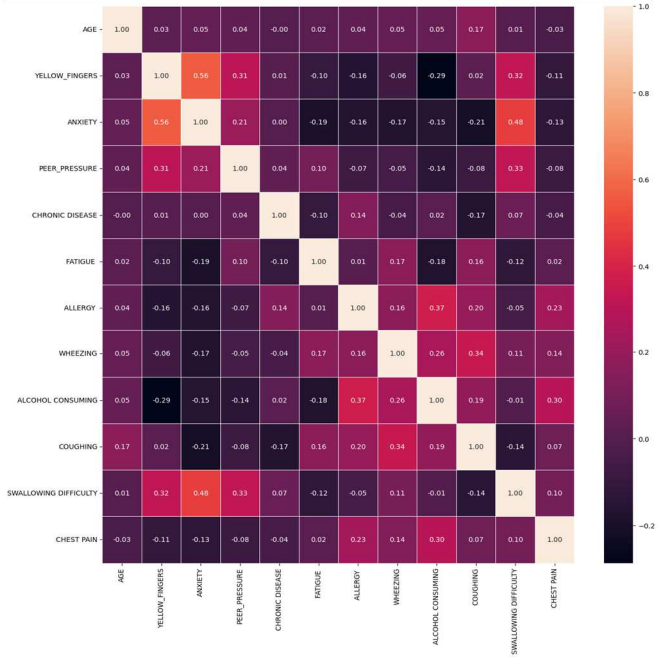


Fig 4: Correlation between all the features

C. *Machine Learning Models*

Throughout the current investigation, specialists used a variety of machine learning techniques to forecast the development of lung cancer. To accomplish the goal, scientists carefully selected the following machine learning models: SVM, Random Forest, KNN, Logistic Regression, Naive Bayes, and Decision Trees, due to their efficacy in medical diagnosis. Prior to putting any machine learning technique into practice, hyperparameter tweaking was done. Hyperparameter tuning was carried by using RandomizedSearchCV. The values of the parameters utilized for every single classifier are displayed in **Table II.**

TABLE II. HYPERPARAMETER VALUES FOR EACH CLASSIFIER

Classifier	Parameter	Value
Support Vector Machine (SVM)	gamma	1
	C	10
Random Forest	n_estimators	50
	max_depth	8
KNN	n_neighbors	1
Logistic Regression	C	10
Gaussian Naive Bayes	No parameter grid needed for Naive Bayes	
Decision Tree	max_depth	6
	min_samples_split	2
	min_samples_leaf	2

During the contagious period, an ill person will be isolated and provided with sufficient support to begin medication as soon as possible. This study examines the predictive accuracy of many AI techniques using machine learning and data on lung cancer.

SVM

A popular machine learning technique that includes prediction and data classification is the support vector machine [**19**]. It maps every information item into a multidimensional feature space and, once the information is classified into two groups, establishes a hypervisor [**19**]. For linear data, it establishes the best hypervisor by reducing the generalize mistakes and the minimal distance within the two groups, such as {positive (+1) and negative (-1)}.

D= $\{(D_1, y_1), (D_2, y_2) \ldots, (D_n, y_n)\}$ denotes the data set being used, which represents the trained example with related positive or negative groups y_i where $y_i \in$ {+1 or -1} [**19**]. Equation (**1**) is used to form a hyperplane that separates by measuring the resulting distance maximization of $W^T D_i + b = -1$ for y_1 = diagnosis negative and $W^T D_i + b = +1$ for y diagnosis positive. Equation (**2**) then shows the classification accuracy for those points meet it [**20**].

$$W^T D_i + b = 0 \quad (1)$$

$$Y_i . W^T D_i + b \geq 1 \quad (2)$$

Here, the optimization function is denoted by $f(\dot{w}, \dot{D}_i)$ such that the value of $\frac{2}{\|w\|}$ [**20**].

$$(\dot{w}, \dot{D}_i) = \min \frac{\|w\|}{2} + C_i . \sum_{i=1}^{k} \lambda_i \quad (10)$$

For better performance, the optimum choice of the kernel parameter λ and regularization parameter C is essential [**20**].

D. *Environmental Setup*

The data was initially gathered, split into training and testing sets, and then randomly shuffled. Particular preparation methods were used before to using the testing sets in order to prevent data leaking and overfitting. The study required a specific configuration to achieve the intended results, and the necessary setup, along with the prerequisites for carrying out the research, is detailed in **Table III**.

TABLE III. SYSTEM'S SPECIFICATIONS

Platform	Google Co-Lab
Language	Python
Libraries	Matplotlib, Scikit-Learn, NumPy, Pandas, and Seaborn
Configuration	
RAM	12.68 GB
CPU	2-core Xeon 2.2 GHz
Session Limit	12 h
Disk Space	71 GB

E. *Performance Metrics:*

Several metrics, including accuracy, precision, recall, and F1-score, were used to evaluate the performance of the classification methods applied to the lung cancer data. In the confusion matrix analysis, the terms T, F, P, and N represent true, false, positive, and negative outcomes, respectively. The total number of correctly categorized healthy points of data, for instance, is represented by TP. Equation (**4**) was used in

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the computation to express the overall accuracy. The proportion of accurately predicted positive cases out of all positive predictions can be calculated using formula (**5**). The F1-Score metric, which assesses the harmonic mean of sensitivity and precision, is determined using formula (6). In machine learning, recall measures a model's capability to detect all relevant instances or true positives. Equation (**7**) provides a mathematical definition of recall [21].

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$F1 - Score = \frac{2 \times TP}{2 \times TP + FP + FN}$$

$$Recall = \frac{TP}{TP + FN}$$

IV. Result Analysis

This article evaluated the performance of six machine learning algorithms in identifying lung cancer: Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), Logistic Regression, Naive Bayes, and Decision Tree. A thorough comparison of the performance of these classifiers based on accuracy, precision, recall, and F1 score is shown in **Table IV**.

The SVM approach outperformed all other models, achieving the maximum accuracy of 99.38%, precision of 99.39%, recall of 99.38%, and F1 score of 99.38%. This enhanced performance demonstrates how SVM has a great deal of potential as a lung cancer screening tool. SVM's extensive performance metrics show how reliable and resilient it is at reliably and low-error rarely recognizing cases of lung cancer.

TABLE IV. Evaluation of the Explainable Aspects of Six Machine Learning Techniques.

Index	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
1	Support Vector Machine	99.38	99.39	99.38	99.38
2	Random Forest	96.30	96.55	96.30	96.29
3	K-Nearest Neighbors	96.91	97.09	96.91	96.91
4	Logistic Regression	88.89	88.91	88.89	88.89
5	Naive Bayes	90.74	90.90	90.74	90.73
6	Decision Tree	92.59	92.62	92.59	92.59

Accuracy rates for KNN and Random Forest were 96.91% and 96.30%, respectively, indicating strong performance. These models proved to be reliable SVM replacements, exhibiting high recall and accuracy scores. Naive Bayes performed moderately, with an accuracy of 92.59% and 90.74 Decision Tree, while Logistic Regression had the lowest accuracy of 88.89%, indicating that it might not be as effective for this particular dataset.

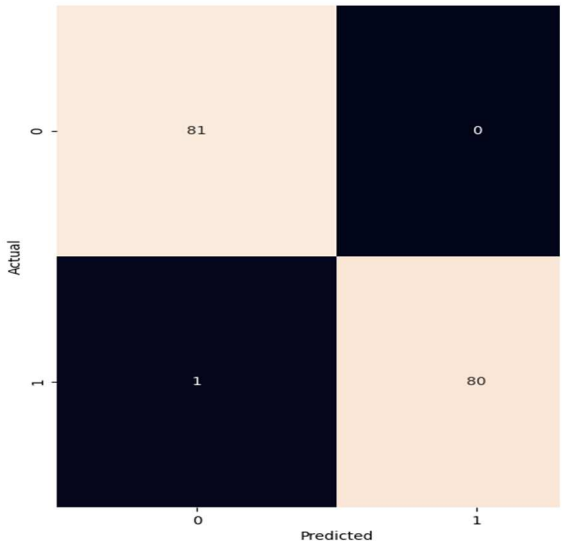


Fig 5. Confusion matrix for SVM

Figure **5** shows the confusion matrix of SVM to predict lung cancer. The true positives, true negatives, false positives, and false negatives are all represented visually by the matrix. The ROC curves in Figure **6** show the specificity versus sensitivity trade-offs for each classifier, with SVM having the most favorable curve.

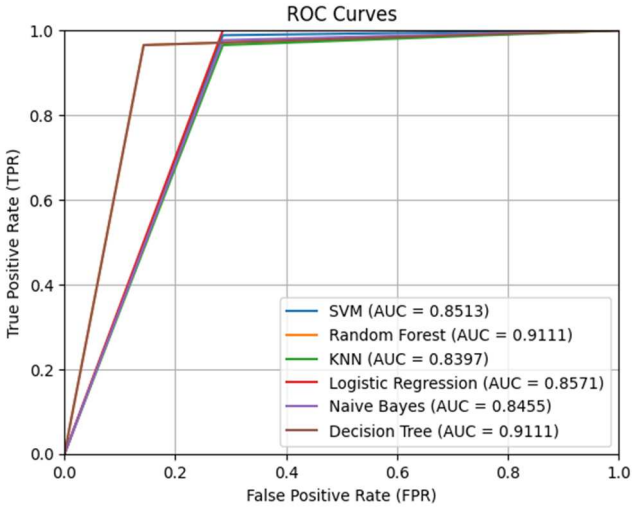


Fig 6: Comparison of the ROC curves for six accuracy models

Table V ultimately summarizes the evaluation of many machine learning algorithms for lung cancer identification. Through sophisticated preprocessing and tuning of the Support Vector Machine (SVM) model, our work achieves the best accuracy (99.38%), surpassing the results of earlier studies utilizing the same lung cancer dataset. Our method dramatically improved model performance by addressing important concerns such as data inconsistencies and class imbalances, which were not addressed in previous researches. Through the use of strategies including feature selection, data encoding, duplicate data removal, and oversampling, we have established a strong model that raises the bar for lung cancer detection research.

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TABLE V. EVALUATING SEVERAL TYPES OF MACHINE LEARNING METHODS TO DETERMINE WHETHER LUNG CANCER EXISTS

Authors	Data modality	Dataset used	No. of Instances	Classification	Accuracy
Olatunji et al. [12]	Clinical data	-	309	Support Vector Machine (SVM)	97.27%
Çelik et al. [17]	Clinical data	-	-	Random Forest	96.08%
Faisal et al. [16]	Clinical data	UCI repository	-	Gradient-boosted Tree	90%
Bhuiyan et al. [15]	Clinical data	Various clinics and hospitals	5000	LightGBM	93.50%
Dritsas et al. [14]	Clinical data	Kaggle online website	309	Rotation Forest	97.10%
Proposed Model	***Clinical data***	***Kaggle online website [18]***	***309***	***Support Vector Machine (SVM)***	***99.38%***

V. CONCLUSION

Our research underscores the efficacy of machine learning, especially the SVM algorithm, in precisely identifying lung cancer by employing thorough data preprocessing and comprehensive model assessment. By feature selection, encoding categorical variables, removing duplicates, and applying oversampling techniques to address class imbalances, we ensured a reliable and robust dataset. Among the algorithms evaluated, SVM delivered outstanding performance with an accuracy of 99.38%. Future efforts will aim to improve model interpretability by leveraging explainable AI techniques such as SHAP (Shapley Additive Explanations) to gain deeper insights into the contributions of individual features. We also aim to explore deep learning approaches to further improve predictive accuracy and reliability, expand the dataset to encompass more diverse patient demographics and larger sample sizes to reinforce model robustness, and integrate medical imaging data to develop multimodal models combining clinical and imaging information for a more holistic diagnostic tool. Ultimately, real-world testing and deployment in clinical environments will be pursued to ensure the models meet the practical needs of patients and healthcare providers.

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